In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Create reproducible population
np.random.seed(42)
n = 300
departments = ["Computer", "Electronics", "Mechanical", "Civil"]
clusters = [f"Class_{i}" for i in range(1, 11)]
genders = ["Male", "Female"]
df = pd.DataFrame({
"Student_ID": range(1, n + 1),
"Department": np.random.choice(
departments, n, p=[0.40, 0.25, 0.20, 0.15]),
"Gender": np.random.choice(
genders, n, p=[0.55, 0.45]),
"Cluster": np.random.choice(clusters, n),
"Age": np.random.randint(18, 24, n),
"Score": np.clip(
np.random.normal(70, 10, n), 0, 100)
})
print("First 10 records:")
print(df.head(10))
print("\nPopulation size:", len(df))
print("Population mean score:", df["Score"].mean())
print("Population standard deviation:", df["Score"].std())
# ====================================================
# A. SIMPLE RANDOM SAMPLING
# ====================================================
srs_sample = df.sample(n=30, random_state=42)
print("\n========== SRS ==========")
print(srs_sample)
print("Sample size:", len(srs_sample))
print("Mean score:", srs_sample["Score"].mean())
print("Standard deviation:", srs_sample["Score"].std())
# ====================================================
# B. STRATIFIED SAMPLING
# ====================================================
stratified_sample = (
df.groupby("Department", group_keys=False)
.apply(lambda g: g.sample(frac=0.10, random_state=42))
.reset_index(drop=True)
)
print("\n========== STRATIFIED ==========")
print(stratified_sample)
print("Sample size:", len(stratified_sample))
print("Mean score:", stratified_sample["Score"].mean())
print("Standard deviation:", stratified_sample["Score"].std())
print("\nPopulation department proportions:")
print(df["Department"].value_counts(normalize=True))
print("\nSample department proportions:")
print(stratified_sample["Department"].value_counts(normalize=True))
# ====================================================
# C. CLUSTER SAMPLING
# ====================================================
```

```
selected_clusters = np.random.choice(
clusters, size=3, replace=False)
cluster_sample = df[
df["Cluster"].isin(selected_clusters)
].copy()
print("\n========== CLUSTER ==========")
print("Selected clusters:", selected_clusters)
print("Sample size:", len(cluster_sample))
print("Mean score:", cluster_sample["Score"].mean())
print("Standard deviation:", cluster_sample["Score"].std())
# ====================================================
# D. COMPARISON
# ====================================================
comparison = pd.DataFrame({
   "Method": ["Population", "SRS", "Stratified", "Cluster"],
   "Sample_Size": [
       len(df), len(srs_sample),
       len(stratified_sample), len(cluster_sample)],
   "Mean_Score": [
       df["Score"].mean(),
       srs_sample["Score"].mean(),
       stratified_sample["Score"].mean(),
       cluster_sample["Score"].mean()],
   "Std_Dev": [
       df["Score"].std(),
       srs_sample["Score"].std(),
       stratified_sample["Score"].std(),
       cluster_sample["Score"].std()]
})
print("\n========== COMPARISON ==========")
print(comparison)
# ====================================================
# E. VISUALIZATION
# ====================================================
plt.figure(figsize=(8, 5))
plt.bar(comparison["Method"], comparison["Mean_Score"])
plt.axhline(
df["Score"].mean(),
linestyle="--",
label="Population Mean")
plt.xlabel("Sampling Method")
plt.ylabel("Mean Score")
plt.title("Population and Sample Mean Comparison")
plt.legend()
plt.show()
```

First 10 records:

	Student_ID	Department	Gender	Cluster	Age	Score
0	1	Computer	Male	Class_5	20	91.910660
1	2	Civil	Male	Class_2	18	73.400676
2	3	Mechanical	Male	Class_10	20	57.259310
3	4	Electronics	Female	Class_6	20	84.140405
4	5	Computer	Female	Class_5	21	61.442253
5	6	Computer	Female	Class_6	20	97.418783
6	7	Computer	Male	Class_1	20	70.796579
7	8	Civil	Male	Class_5	23	80.133160
8	9	Electronics	Female	Class_9	18	49.199824
9	10	Mechanical	Male	Class_10	21	57.158591

Population size: 300
Population mean score: 71.219789651472
Population standard deviation: 10.976831392461916

========== SRS ==========

	Student_ID	Department	Gender	Cluster	Age	Score
203	204	Civil	Female	Class_10	22	85.673530
266	267	Mechanical	Male	Class_6	23	62.330468
152	153	Computer	Male	Class_5	22	86.381127
9	10	Mechanical	Male	Class_10	21	57.158591
233	234	Electronics	Female	Class_3	22	66.268348
226	227	Civil	Male	Class_4	20	78.737830
196	197	Mechanical	Female	Class_3	20	83.514784
109	110	Computer	Male	Class_9	22	74.917931
5	6	Computer	Female	Class_6	20	97.418783
175	176	Computer	Female	Class_9	20	66.168175
237	238	Computer	Male	Class_5	18	81.335468
57	58	Computer	Female	Class_9	23	71.773754
218	219	Electronics	Female	Class_10	21	56.553146
45	46	Mechanical	Female	Class_7	20	63.405866
182	183	Civil	Male	Class_6	18	76.321161
221	222	Electronics	Male	Class_1	20	73.517547
289	290	Electronics	Male	Class_10	22	76.292490
211	212	Mechanical	Male	Class_1	23	80.427652
148	149	Computer	Male	Class_6	23	86.265195
165	166	Mechanical	Male	Class_5	18	69.277165
78	79	Computer	Female	Class_6	23	81.037139
113	114	Mechanical	Female	Class_3	19	88.468060
249	250	Civil	Female	Class_9	20	73.785689
250	251	Computer	Male	Class_1	23	68.926740
104	105	Civil	Male	Class_6	22	66.488593
42	43	Computer	Male	Class_3	22	78.315428
281	282	Computer	Female	Class_5	19	71.777559
295	296	Electronics	Male	Class_6	19	67.868647
157	158	Mechanical	Male	Class_2	23	65.356089
238	239	Electronics	Male	Class_4	18	57.227795

Sample size: 30
Mean score: 73.76635830118529
Standard deviation: 10.085271977921481

========== STRATIFIED ==========

	Student_ID	Department	Gender	Cluster	Age	Score
0	192	Civil	Female	Class_5	19	75.593592

```
1          250        Civil  Female   Class_9   20   73.785689
2          184        Civil  Female   Class_1   22   60.212334
3          262        Civil    Male   Class_9   18   67.047378
4          179        Civil  Female   Class_8   20   70.625027
5          124     Computer    Male   Class_1   22   77.312155
6           84     Computer  Female   Class_6   18   75.467794
7           41     Computer    Male  Class_10   22   75.630789
8           65     Computer  Female   Class_6   19   68.190093
9          260     Computer  Female   Class_1   18   80.730522
10         125     Computer    Male   Class_1   21   76.782994
11         152     Computer  Female   Class_1   19   67.431897
12         237     Computer  Female  Class_10   18   75.997839
13         178     Computer  Female   Class_8   23   58.535088
14          59     Computer  Female   Class_4   20   51.691846
15         210     Computer  Female   Class_6   20   67.959694
16          60     Computer  Female  Class_10   21  100.000000
17         144     Computer    Male   Class_5   18   87.208303
18         211  Electronics  Female   Class_9   20   68.139435
19          88  Electronics    Male   Class_7   23   62.574856
20          21  Electronics    Male  Class_10   22   65.623343
21          40  Electronics    Male   Class_2   23   67.248760
22         130  Electronics    Male   Class_3   21   65.052966
23         188  Electronics    Male   Class_1   19   83.831450
24         264  Electronics    Male   Class_3   18   69.062346
25         186   Mechanical  Female   Class_1   22   71.327807
26          39   Mechanical    Male   Class_7   23   60.465408
27         187   Mechanical    Male   Class_9   18   87.016962
28          76   Mechanical    Male  Class_10   20   70.358945
29         108   Mechanical    Male   Class_6   19   62.961053
30         272   Mechanical  Female  Class_10   23   50.478148
Sample size: 31
Mean score: 70.78530685429115
Standard deviation: 10.384933673951592

Population department proportions:
Department
Computer       0.430000
Electronics    0.226667
Mechanical     0.183333
Civil          0.160000
Name: proportion, dtype: float64

Sample department proportions:
Department
Computer       0.419355
Electronics    0.225806
Mechanical     0.193548
Civil          0.161290
Name: proportion, dtype: float64

========== CLUSTER ==========
Selected clusters: ['Class_1' 'Class_3' 'Class_2']
Sample size: 80
Mean score: 71.0875505006151
Standard deviation: 11.075979961867777
```

```
========== COMPARISON ==========
       Method  Sample_Size  Mean_Score    Std_Dev
0  Population          300   71.219790  10.976831
1         SRS           30   73.766358  10.085272
2  Stratified           31   70.785307  10.384934
3     Cluster           80   71.087551  11.075980
```

```
/tmp/ipykernel_4455/3265390791.py:40: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.
  .apply(lambda g: g.sample(frac=0.10, random_state=42))
```

